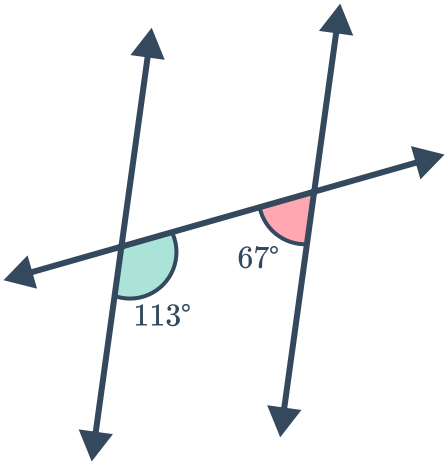
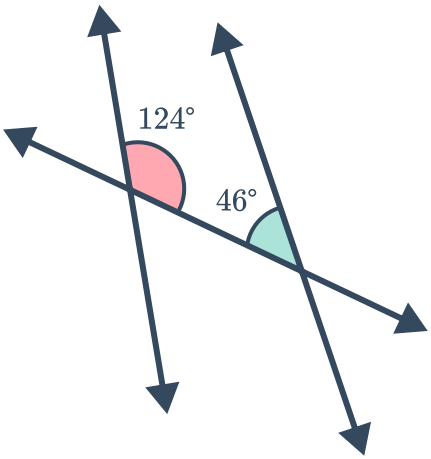
Proving lines parallel

1 For the following, state whether the diagram contains parallel lines. Justify your choice with a theorem or postulate.

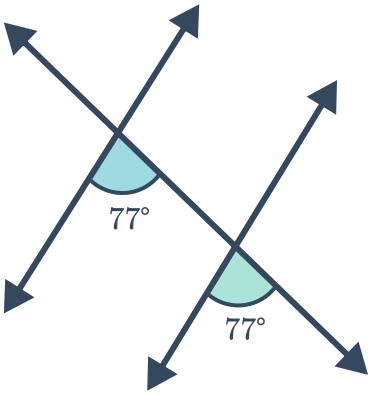
a



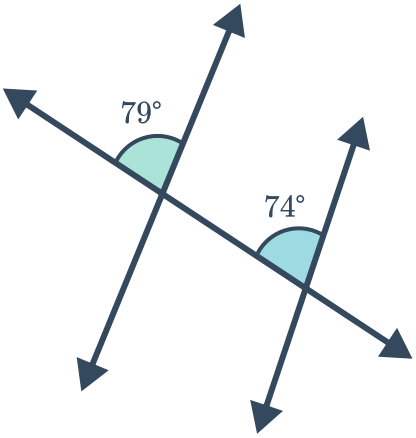
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c

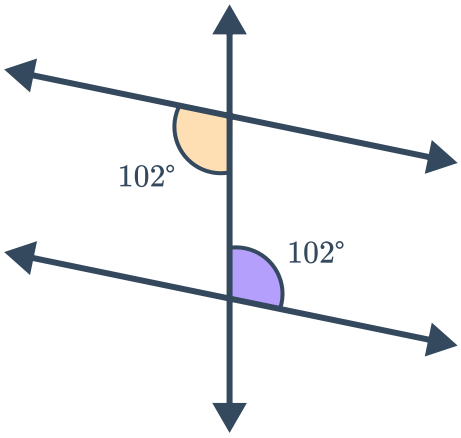


d

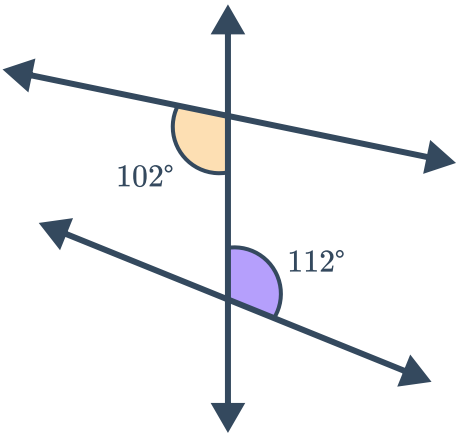


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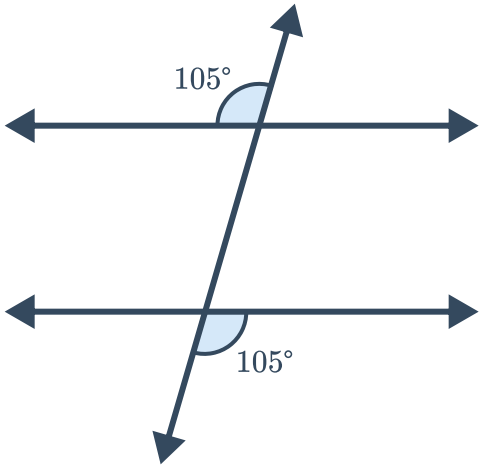
e



f

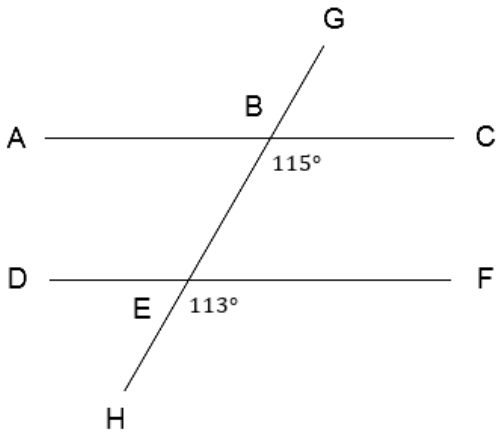


g

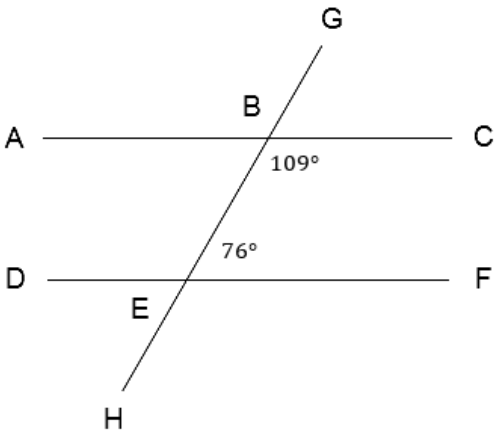


2 For the following diagrams, state whether $\overline{AC}$ and $\overline{DF}$ are parallel. Explain your reasoning.

a



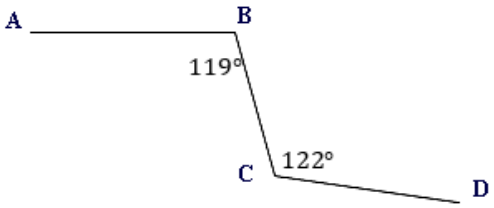
b



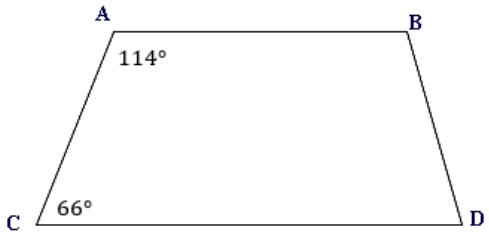
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3 For the following diagrams, state whether $\overline{AB}$ and $\overline{CD}$ are parallel. Explain your reasoning.

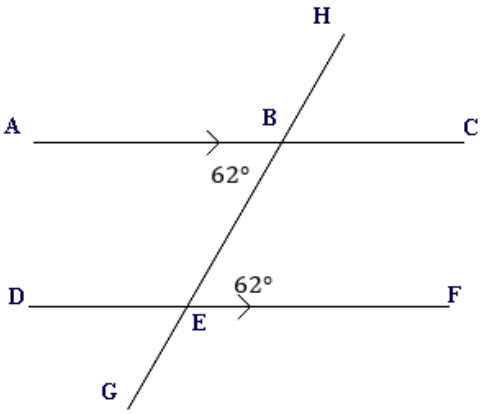
a



b



4 Eileen was given the following diagram, in which $m\angle ABE = m\angle BEF = 62^\circ$. She immediately marked the parallel symbols on the diagram to show that $\overline{AB} \parallel \overline{DE}$. State whether the following reasons justify Eileen's markings.



a The two congruent angles form a pair of corresponding angles. Since the corresponding angles are congruent, $\overline{AB}$ must be parallel to $\overline{DE}$.

b The diagram is drawn to scale and so the lines look to be parallel.

c The two congruent angles form a pair of alternate interior angles. Since the alternate interior angles are congruent, $\overline{AB}$ must be parallel to $\overline{DE}$.

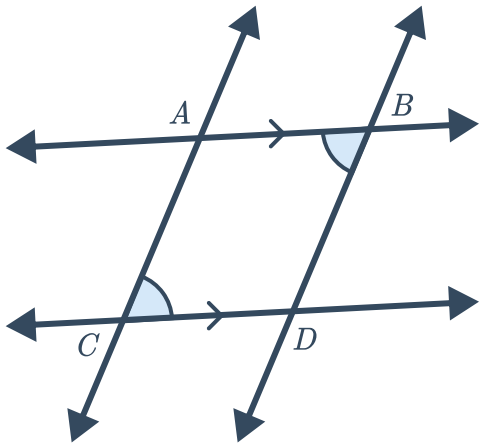
d The two congruent angles form a pair of consecutive interior angles. Since the consecutive interior angles are congruent, $\overline{AB}$ must be parallel to $\overline{DE}$.

5 Suppose that two lines, $\overleftrightarrow{AB}$ and $\overleftrightarrow{CD}$, are intersected by a transversal.
Which of the following reasons alone could be used to show that the lines are parallel?

- Consecutive interior angles converse
- Alternate exterior angles theorem
- Linear pair theorem
- Congruent supplements theorem

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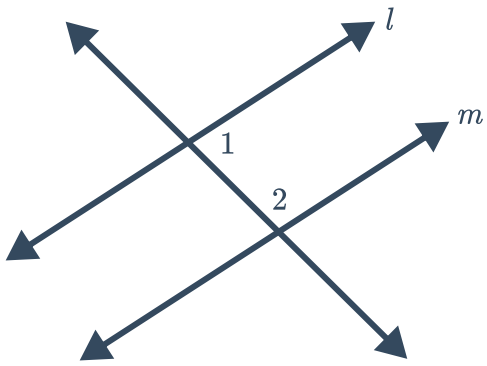
6 The following diagram shows that $\overleftrightarrow{AB} \parallel \overleftrightarrow{CD}$ and $\angle ABD \cong \angle DCA$. Prove that $\overleftrightarrow{AC} \parallel \overleftrightarrow{BD}$.



Additional questions

7 Review the statement. Describe the error.

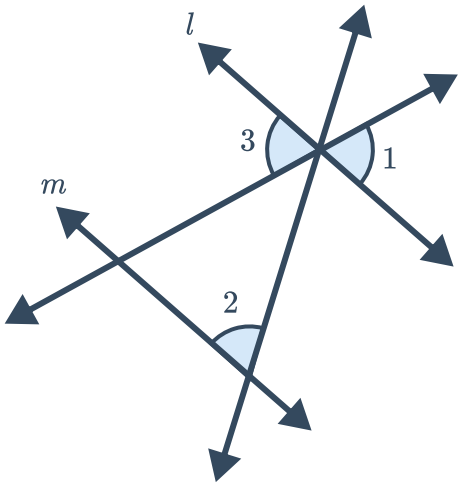
- If $\angle 1 \cong \angle 2$, then by the converse of the consecutive interior angles theorem, $l \parallel m$.



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8 Review the following proof. Describe the error.

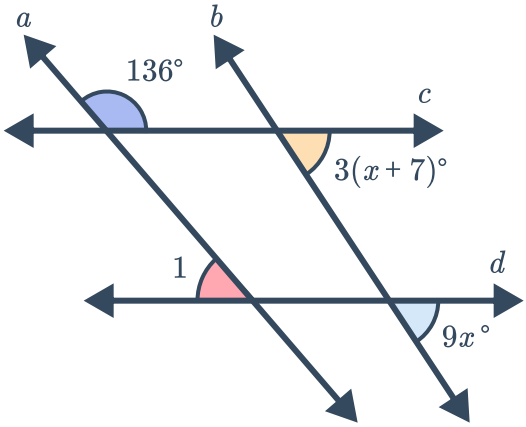
$\angle 1 \cong \angle 2$ is given. By the vertical angles theorem, $\angle 1 \cong \angle 3$. So by the transitive property of congruence, $\angle 2 \cong \angle 3$. By the converse of the corresponding angles theorem, $l \parallel m$



9 Consider the given diagram:

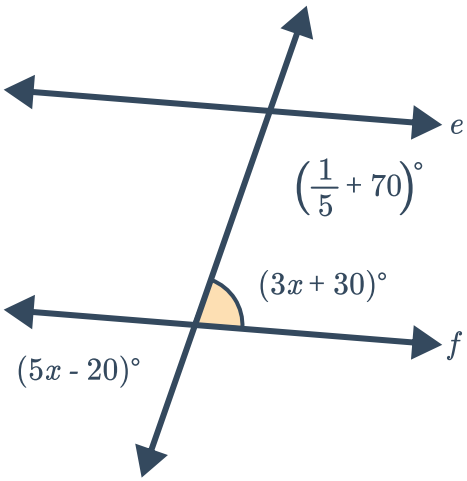
a Find the value of x such that $c \parallel d$.

b Find $m\angle 1$ such that $c \parallel d$. Justify your reasoning.

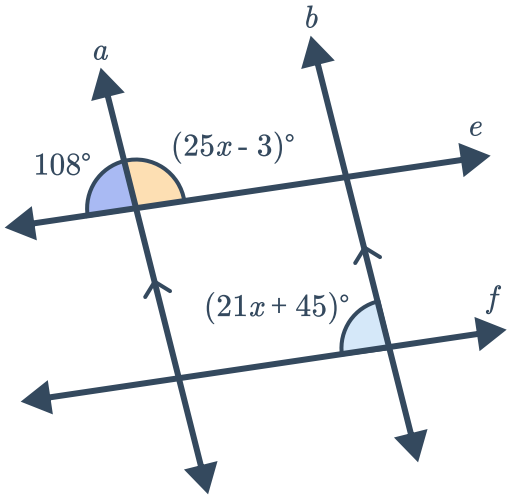


10 Determine the value of x that makes $e \parallel f$.

a

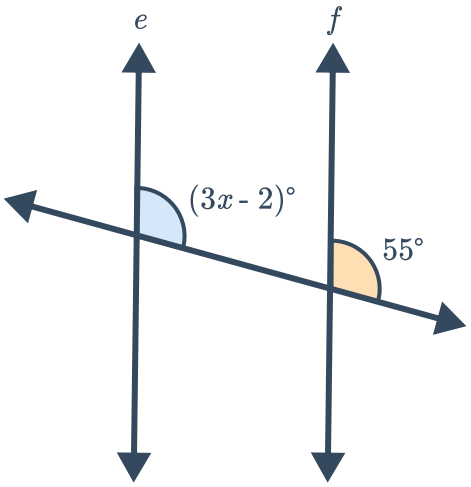


b

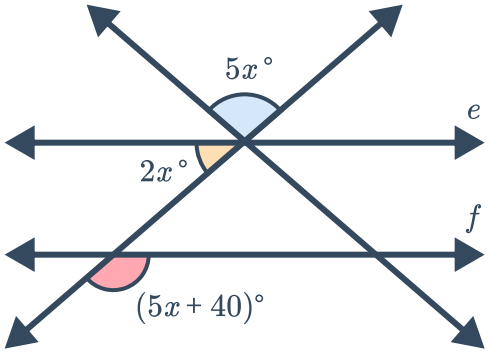


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c

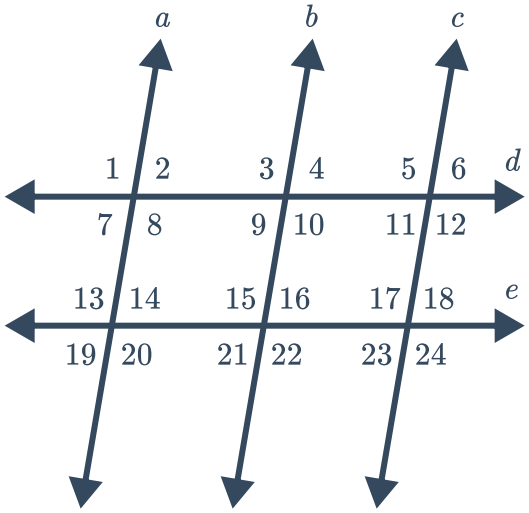


d



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11 Consider the following diagram for the below statements. Determine if the information given is enough to justify the conclusion. Explain your reasoning.



a Given:

- $\angle 1$ and $\angle 14$ are supplementary angles
- $\angle 22 \cong \angle 1$

Conclusion: $a \parallel b$

b Given:

- $\angle 7$ and $\angle 10$ are supplementary

Conclusion: $b \parallel c$

c Given:

- $b \parallel c$
- $\angle 5$ and $\angle 19$ are supplementary

Conclusion: $a \parallel b$

d Given:

- $\angle 1$ and $\angle 11$ are supplementary

Conclusion: $a \parallel c$

e Given:

- $\angle 18$ and $\angle 20$ are supplementary
- $a \parallel b$

Conclusion: $b \parallel c$

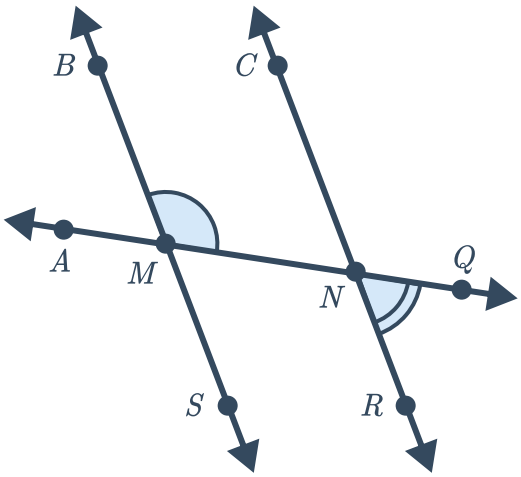
f Given:

- $\angle 1 \cong \angle 13$
- $\angle 3 \cong \angle 24$

Conclusion: $a \parallel b$

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12 Consider the following diagram:



a Identify the angles that are supplementary to $\angle BMN$.

b If $\angle BMN$ and $\angle QNR$ are supplementary, prove $\overleftrightarrow{BS} \parallel \overleftrightarrow{CR}$.

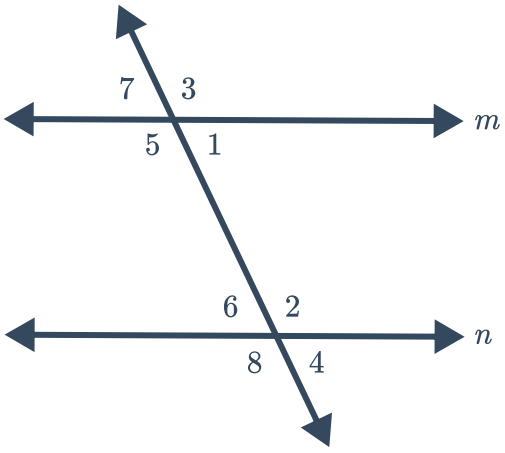
13 Given:

- $\angle A$ and $\angle D$ are supplementary
- $\angle B \cong \angle D$

Prove: $\overline{AD} \parallel \overline{BC}$

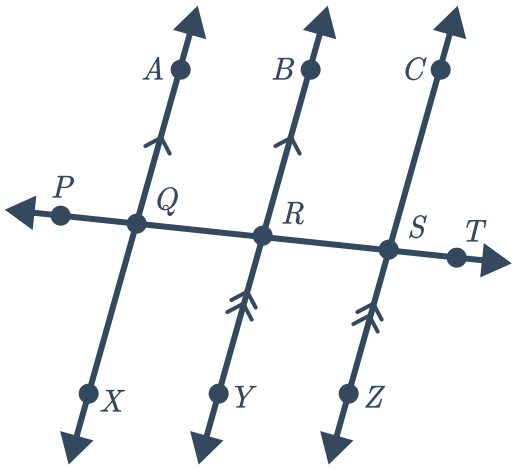


14 Prove the converse of the consecutive interior angles theorem:
Given: $\angle 1$ is supplementary to $\angle 2$
Prove: $m \parallel n$

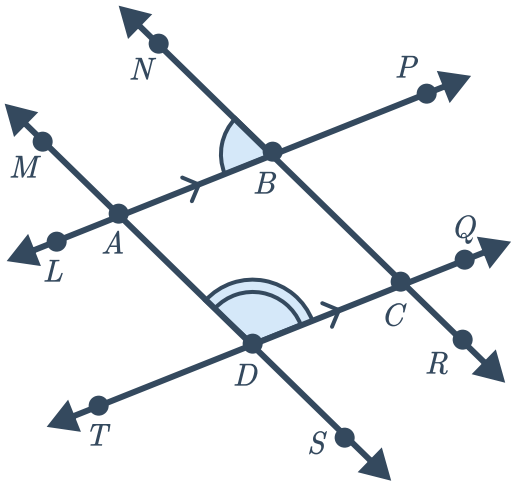


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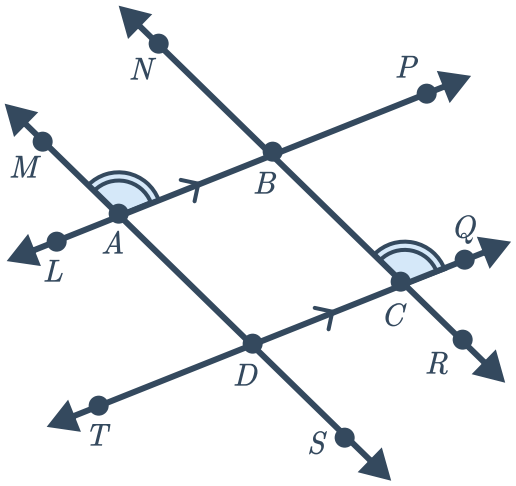
15 Given: $\overleftrightarrow{AX} \parallel \overleftrightarrow{BY}$ and $\overleftrightarrow{BY} \parallel \overleftrightarrow{CZ}$
Prove: $\angle CST \cong \angle XQP$



16 Given: $\overleftrightarrow{AB} \parallel \overleftrightarrow{DC}$ and $\angle NBA$ and $\angle ADC$ are supplementary
Prove: $\overleftrightarrow{AD} \parallel \overleftrightarrow{BC}$



17 Given: $\overleftrightarrow{AB} \parallel \overleftrightarrow{DC}$ and $\angle MAB \cong \angle BCQ$
Prove: $\overleftrightarrow{AD} \parallel \overleftrightarrow{BC}$



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